When $t = 0$, $I = 0$ and when $t \to \infty$, $I = I_0$.

Greater the value of L/R, longer is the time taken by the current I to reach its maximum value (Fig. 12.2).

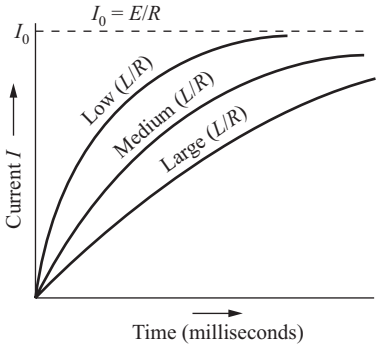


FIG. 12.2

12.2 DECAY OF CURRENT IN A CIRCUIT CONTAINING L AND R

When the circuit is broken, an induced emf, equal to $-L\dfrac{dI}{dt}$ is again produced in the inductance L and it slows down the rate of decay of the current. The current in the circuit decays from maximum value I_0 to zero. During the decay, let I be the current at time t. In this case $E = 0$. The emf equation for the decay of current is

$$0 = RI + L\frac{dI}{dt} \qquad \ldots(1)$$

$$\therefore \quad \frac{dI}{I} = -\frac{R}{L}dt$$

Integrating, $\log_e I = -\dfrac{R}{L}t + C$ where C is a constant

When $t = 0$, $I = I_0$ $\quad \therefore \quad \log_e I_0 = C$

$$\log_e I = -\frac{R}{L}t + \log_e I_0, \quad \text{or} \quad \log_e \frac{I}{I_0} = -\frac{R}{L}t$$

$$\frac{I}{I_0} = e^{-(R/L)t}$$

$$\therefore \quad I = I_0 e^{-(R/L)t} \qquad \ldots(2)$$

Eq. (2) represents the current at any instant t during decay. A graph between current and time is shown in Fig. 12.3.

When $t = L/R$, $I = I_0 e^{-1} = \dfrac{1}{e} I_0 = 0.365\, I_0$

$t = 2L/R$, $I = I_0 e^{-2} = 0.1035\, I_0$

$t = 3L/R$, $I = I_0 e^{-3} = 0.05\, I_0$

Therefore, *the time constant L/R of a R–L circuit may also be defined as the time in which the current in the circuit falls to 1/e of its maximum value* when external source of e.m.f. is removed.

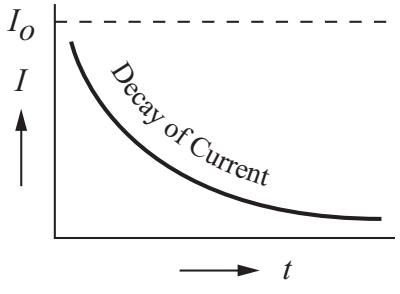


FIG. 12.3

The rate of decay of current is

$$\frac{dI}{dt} = -\frac{R}{L} I_0 e^{-(R/L)t} = -\frac{R}{L} I$$

Thus it is clear that greater the ratio *R/L, or smaller the time constant L/R, the more rapidly does the current die away* (Fig. 12.4).

Fig. 12.5 shows that the growth and decay curves are complementary.

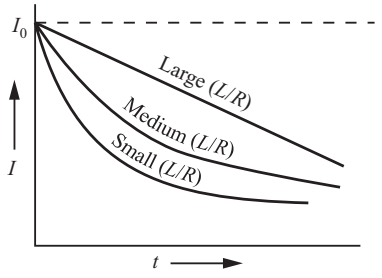


FIG. 12.4

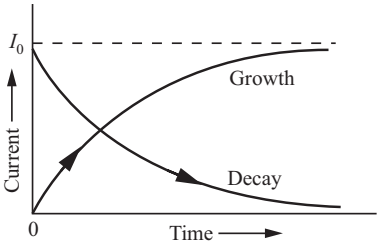


FIG. 12.5

Example 1. *An e.m.f. 10 volts is applied to a circuit having a resistance of 10 ohms and an inductance of 0.5 henry. Find the time required by the current to attain 63.2% of its final value. What is the time constant of the circuit?*

Solution. $I = I_0(1 - e^{-Rt/L})$

Given: $\frac{I}{I_0} = \frac{63.2}{100}; \frac{R}{L} = \frac{10}{0.5} = 20$

$$\frac{63.2}{100} = 1 - e^{-20t}$$

$$e^{-20t} = 1 - 0.632 = 0.368$$

$$e^{20t} = \frac{1}{0.368} = 2.717$$

$$20t = \log_e 2.717$$

$$t = \frac{1}{20} \times 2.3026 \times \log_{10} 2.717$$

$$= \frac{2.3026 \times 0.4341}{20} = \mathbf{0.05\ sec.}$$

The time constant of the circuit is

$$\frac{L}{R} = \frac{0.5}{10} = \frac{1}{20} \text{ sec.}$$

Example 2. *An inductance of 500 mH and a resistance of 5 ohms are connected in series with an e.m.f of 10 volts. Find the final current. If now the cell is removed and the two terminals are connected together, find the current after (i) 0.05 sec. and (ii) 0.2 sec.*

Solution. Final current $I_0 = \frac{E}{R} = \frac{10}{5} = 2A$

During discharge, $I = I_0 e^{-Rt/L}$

Now $\frac{R}{L} = \frac{5}{500 \times 10^{-3}} = 10$

$$\therefore \quad Q = Q_0\left(1 - e^{-\frac{t}{CR}}\right) \qquad ...(3)$$

The term CR is called time constant of the circuit.

At the end of time $t = CR, Q = Q_0\,(1 - e^{-1}) = 0.632\,Q_0$

Thus, *the time constant may be defined as the time taken by the capacitor to get charged to 0.632 times its maximum value.*

The growth of charge is shown in Fig. 12.7.

The rate of growth of charge is

$$\frac{dQ}{dt} = \frac{Q_0}{CR}\,e^{-\frac{t}{CR}} = \frac{I}{CR}\,(Q_0 - Q)$$

Thus it is seen that smaller the product CR, the more rapidly does the charge grow on the capacitor.

The rate of growth of the charge is rapid in the beginning and it becomes less and less as the charge approaches nearer and nearer the steady value.

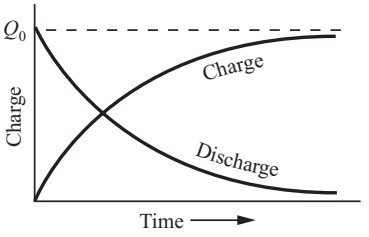


FIG. 12.7

(*b*) Decay of charge (Discharging of a Capacitor through Resistance)

Let the capacitor having charge Q_0 be now discharged by releasing the Morse key K (Fig. 12.6). The charge flows out of the capacitor and this constitutes a current. In this case $E = 0$.

$$R\,\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad \ldots(1)$$

or $$\frac{dQ}{Q} = -\frac{1}{CR}\,dt$$

Integrating, $\log_e Q = -\dfrac{t}{CR} + K,$ where K is a constant

When $t = 0, Q = Q_0 \therefore \log_e Q_0 = K$

$$\log_e Q = -\frac{t}{CR} + \log_e Q_0$$

or $$\log_e \frac{Q}{Q_0} = -\frac{t}{CR} \quad \text{or} \quad \frac{Q}{Q_0} = e^{-t/CR}$$

$$\therefore \quad Q = Q_0 e^{-t/CR} \qquad \ldots(2)$$

This shows that the charge in the capacitor decays exponentially and becomes zero after infinite interval of time (Fig. 12.7).

The rate of discharge is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR}\,e^{-t/CR} = -\frac{Q}{CR} \qquad \ldots(3)$$

Thus, smaller the time-constant CR, the quicker is the discharge of the capacitor.

In Eq. (2), if we put $t = CR$, then $Q = Q_0 e^{-1} = 0.368\,Q_0$.

Hence *time constant may also be defined as the time taken by the current to fall from maximum to 0.368 of its maximum value.*

Example 1. *A capacitor is charged by DC supply through a resistance of 2 megaohms. If it takes 0.5 seconds for the charge to reach three quarters of its final value, what is the capacitance of the capacitor?*

Solution. Here, $R = 2 \times 10^6\ \Omega$, $t = 0.5$s, $Q/Q_0 = 3/4$.

$$Q = Q_0\,(1 - e^{-t/CR}) \quad \text{or} \quad \frac{Q}{Q_0} = (1 - e^{-t/CR})$$

or $$\frac{3}{4} = \left(1 - e^{-\frac{0.5}{C\times(2\times10^6)}}\right)$$

or $$e^{\frac{0.5}{C\times(2\times10^6)}} = 4 \quad \text{or} \quad \frac{0.5}{C \times (2\times10^6)} = \log_e 4$$

or $$C = \frac{0.5}{(2.3026 \log_{10} 4)\,(2\times10^6)} = 0.18\times10^{-6}\ \text{F} = 0.18\ \mu\text{F}.$$

Example 2. *A capacitor of capacitance 0.1 μF is first charged and then discharged through a resistance of 10 megaohm. Find the time, the potential will take to fall to half its original value.*

Solution. $Q = Q_0\ e^{-t/CR}$ or $\frac{Q}{Q_0} = e^{-t/CR}$ or $\ln\left(\frac{Q_0}{Q}\right) = \frac{t}{CR}$

$\therefore$ $$t = CR \ln\left(\frac{Q_0}{Q}\right)$$

$Q = CV$ and $Q_0 = CV_0$. $\therefore$ $\frac{Q_0}{Q} = \frac{V_0}{V}$

$\therefore$ $$t = CR \ln\left(\frac{V_0}{V}\right)$$

Here, $C = 10^{-7}\ F$; $R = 10^7\ \Omega$, $V_0/V = 2$.

$\therefore$ $t = 10^{-7} \times 10^7 \times \ln 2 = 0.6931$ s.

Example 3. A *resistance R and a 2μF capacitor in series are connected to a 200 volt direct supply. Across the capacitor is a neon lamp that strikes at 120 volts. Calculate the value of R to make the lamp strike 5 seconds after switch has been closed.*

Solution. The resistance R must be such that the p.d. across the capacitor should rise to 120 volt in 5 seconds after the switch is closed. The lamp would then strike,

The equation of charging is

$$Q = Q_0(1 - e^{-t/RC})$$

or $$CV = CV_0(1 - e^{-t/RC})$$

or $$V = V_0\,(1 - e^{-t/RC})$$

Here, $V = 120$ volt, $V_0 = 200$ volt, $t = 5$ sec and

$C = 2 \times 10^{-6}$ farad,

$\therefore$ $$120 = 200\,(1 - e^{-5/2\times10^{-6}R})$$

or $$e^{5/2\times10^{-6}R} = \frac{5}{2}$$

$$\frac{5}{2\times10^{-6}R} = \log_e 5 - \log_e 2 = 1.6094 - 0.6931 = 0.9163$$

or $$R = \frac{5}{2\times10^{-6}\times0.9163} = 2.73\times10^6\ \text{ohm} = \mathbf{2.73\ megaohm}$$

12.7 DISCHARGE OF A CAPACITOR THROUGH AN INDUCTOR AND A RESISTOR IN SERIES (DECAY OF CHARGE IN LCR CIRCUIT)

Consider a circuit containing a capacitor of capacitance C, an inductance L and a resistance R joined in series (Fig. 12.15). E is a cell. K_2 is kept open. The capacitor is charged to maximum charge Q_0 by closing the key K_1. On opening K_1 and closing key K_2, the capacitor discharges through the inductance L and resistance R. Let I be the current in the circuit and Q be the charge in the capacitor at any instant during discharge. The circuit equation then is

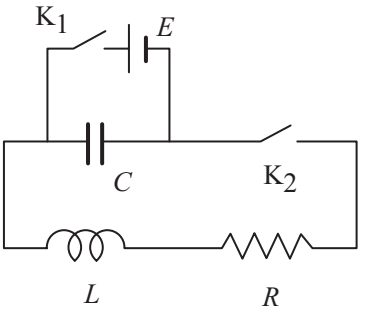


FIG. 12.15

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = 0$$

But, $I = \dfrac{dQ}{dt}$ and $\dfrac{dI}{dt} = \dfrac{d^2Q}{dt^2}$

$\therefore$ $$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = 0$$

$$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{Q}{LC} = 0 \quad \ldots(1)$$

Let $\dfrac{R}{L} = 2b$ and $\dfrac{1}{LC} = k^2$, then

$$\frac{d^2Q}{dt^2} + 2b\frac{dQ}{dt} + k^2Q = 0 \quad \ldots(2)$$

The general solution of this equation is

$$Q = A\,e^{(-b+\sqrt{b^2-k^2})t} + Be^{(-b-\sqrt{b^2-k^2})t} \quad \ldots(3)$$

where A and B are arbitrary constants.

When $t = 0$, $Q = Q_0$ and from Eq. (3), $A + B = Q_0$ …(4)

$$\frac{dQ}{dt} = A\,(-b+\sqrt{b^2-k^2})\,e^{(-b+\sqrt{b^2-k^2})t} + B\,(-b-\sqrt{b^2-k^2})\,e^{(-b-\sqrt{b^2-k^2})t}$$

When, $t = 0$, $\dfrac{dQ}{dt} = 0$

$\therefore$ $$A\,(-b+\sqrt{b^2-k^2}) + B\,(-b-\sqrt{b^2-k^2}) = 0$$

$$-b\,(A+B) + \sqrt{b^2-k^2}\,(A-B) = 0$$

$$-bQ_0 + \sqrt{b^2-k^2}\,(A-B) = 0$$

$\therefore$ $$A - B = \frac{bQ_0}{\sqrt{b^2-k^2}} \quad \ldots(5)$$

From Eqs. (4) and (5), we get

$$A = \frac{1}{2}Q_0\left(1 + \frac{b}{\sqrt{(b^2-k^2)}}\right) \text{ and } B = \frac{1}{2}Q_0\left(1 - \frac{b}{\sqrt{(b^2-k^2)}}\right)$$

Putting these values of A and B in Eq. (3), we get

$$Q = \frac{1}{2} Q_0 e^{-bt} \left[\left(1 + \frac{b}{\sqrt{(b^2 - k^2)}} \right) e^{\sqrt{(b^2 - k^2)}\, t} + \left(1 - \frac{b}{\sqrt{(b^2 - k^2)}} \right) e^{-\sqrt{(b^2 - k^2)}\, t} \right] \quad \ldots(6)$$

Case I. If $b^2 > k^2, \sqrt{(b^2 - k^2)}$ is real and positive and the charge of the capacitor decays exponentially, becoming zero asymptotically (curve 1 of Fig. 12.16). This discharge is known as *over damped, non-oscillatory or dead beat*.

Case II. When $b^2 = k^2$, $Q = Q_0 (1 + bt)\, e^{-bt}$

This represents a *non-oscillatory* discharge. This discharge is known as *critically damped* (Curve 2 of Fig. 12.16). The charge decreases to zero exponentially in a short time.

Case III. If $b^2 < k^2, \sqrt{(b^2 - k^2)}$ is imaginary.

$$\sqrt{(b^2 - k^2)} = i\omega, \qquad \text{where } \omega = \sqrt{(k^2 - b^2)}$$

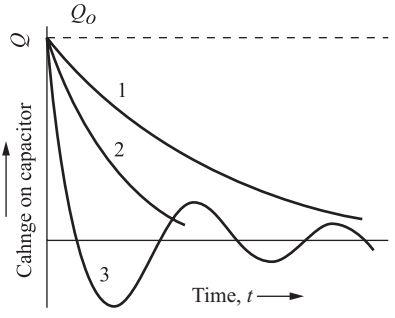


FIG. 12.16

$$\therefore \quad Q = \frac{1}{2} Q_0\, e^{-bt} \left[\left(1 + \frac{b}{i\omega} \right) e^{i\omega t} + \left(1 - \frac{b}{i\omega} \right) e^{-i\omega t} \right]$$

$$= Q_0 e^{-bt} \left[\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) + \frac{b}{\omega} \left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right) \right]$$

$$= \frac{Q_0 e^{-bt}}{\omega} (\omega \cos \omega t + b \sin \omega t)$$

Let $\omega = k \sin \alpha$ and $b = k \cos \alpha$ so that $\tan \alpha = \dfrac{\omega}{b}$.

$$Q = \frac{Q_0 e^{-bt} k}{\omega} (\cos \omega t \sin \alpha + \cos \alpha \sin \omega t)$$

$$= \frac{Q_0 e^{-bt} k}{\omega} \sin (\omega t + \alpha)$$

$$Q = \frac{Q_0 e^{-\frac{R}{2L} t}}{\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2} \right) \sqrt{LC}}} \sin \left(\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2} \right)} t + \alpha \right) \quad \ldots(7)$$

This equation represents a *damped oscillatory* charge as shown by the curve (3). The charge oscillates above and below zero till it finally settles down to zero value.

The frequency of oscillation in the circuit is given by

$$v = \frac{\omega}{2\pi} = \frac{\sqrt{k^2 - b^2}}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

When $R = 0$, $v = \dfrac{1}{2\pi\sqrt{LC}}$

The condition for oscillatory discharge is

$$\frac{R^2}{4L^2} < \frac{1}{LC} \quad \text{or} \quad R < 2\sqrt{\frac{L}{C}}$$

Importance in Wireless Telegraphy. The discharge of a capacitor through an inductance is oscillatory if the resistance R of the circuit is less than $2\sqrt{(L/C)}$. During the discharge, the energy of the charged capacitor is stored in the magnetic field produced in the inductance coil, then again back in the electric field between the capacitor plates, and so on. If the fields are caused to alternate rapidly, some energy escapes from the circuit permanently in the form of electro-magnetic waves which travel through space with the speed of light. These waves form the basis of wireless telegraphy.

Messages can be transmitted from one place to another with the help of codes.

Example 1. *If a battery, of emf 100 volts, is connected in series with an inductance of 10 mH, a capacitor of 0.05* μ*F and a resistance of 100* Ω*, find (i) the frequency of the oscillatory current, (ii) the logarithmic decrement and (iii) the final capacitor charge.*

Solution. $\dfrac{R^2}{4L^2} = \dfrac{100^2}{4 \times (10^{-2})^2} = 2.5 \times 10^7$

and $\dfrac{1}{LC} = \dfrac{1}{10^{-2} \times 0.05 \times 10^{-6}} = 2 \times 10^9$

$R^2/4L^2 < 1/LC$. *Hence the charging of the capacitor is oscillatory.*

$$v = \frac{1}{2\pi}\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)} = \frac{1}{2 \times 3.14}\sqrt{(2 \times 10^9 - 2.5 \times 10^7)} = 7076.6 \text{ Hz.}$$

The logarithmic decrement $= \dfrac{RT}{2L} = \dfrac{100}{2 \times 10^{-2} \times 7076.6} = 0.707.$

The final capacitor charge $Q_0 = EC = 100 \times 0.05 \times 10^{-6} = 5\mu C$.

Example 2. *A charged capacitor of capacitance 0.01* μ*F is made to discharge through a circuit consisting of a coil of inductance 0.1 henry and an unknown resistance. What should be the maximum value of the unknown resistance, if the discharge of the capacitor is to be oscillatory?*

Sol. If R is the maximum value of the resistance for the discharge to be oscillatory, then

$$\frac{R^2}{4L^2} = \frac{1}{LC} \text{ or } R = \sqrt{\frac{4L}{C}} = \sqrt{\left(\frac{4 \times 0.1}{0.01 \times 10^{-6}}\right)} = 6324\,\Omega$$

Example 3. *(i) Find out whether the discharge of a capacitor through a circuit containing the following elements, is oscillatory.*

$$C = 0.2\ \mu F, \quad L = 10\ mH, \quad R = 250\ ohm$$

(*ii*) *If so, find the frequency.*

(*iii*) *Calculate the maximum value of the resistance possible so as to make the discharge oscillatory.*

Solution. (*i*) The condition for oscillations is that

$$\frac{R^2}{4L^2} < \frac{1}{LC}$$

or $$R^2 < \frac{4L}{C}, \quad \text{or} \quad R < \sqrt{\frac{4L}{C}}$$

We have $$\sqrt{\frac{4L}{C}} = \sqrt{\frac{4\times10\times10^{-3}}{0.2\times10^{-6}}} = 447 \text{ ohm}$$

It is given that $R = 250$ ohm

$\therefore$ $$R < \sqrt{\frac{4L}{C}}$$

Therefore the discharge is oscillatory.

(*ii*) The frequency of oscillations is given by

$$v = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{1}{10\times10^{-3}\times0.2\times10^{-6}} - \frac{(250)^2}{4\times(10\times10^{-3})^2}}$$

$$= 54\times10^6 \text{ Hz.}$$

(*iii*) The maximum possible resistance of oscillatory discharge is given by the equation

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

$$R = \sqrt{\frac{4L}{C}} = 447 \text{ ohm.}$$

EXERCISE

1. The inductive time constant is:

(*a*) $\frac{L}{R}$ (*b*) T (*c*) CR (*d*) LR

(Kumaun 2015)

2. If $C = 0.1$ μF and $R = 100$ mega ohm, the time constant for the circuit will be:

(*a*) 1 sec (*b*) 10 sec (*c*) 100 sec (*d*) 1000 sec

(Kumaun 2011, 2014)

3. Derive an expression for growth of current through inductance L and resistance R in a circuit.

(Kumaun 2014)

4. Analyse the growth and decay of current in an *LR* circuit. **(Kumaun 2011)**

5. Define time constant of *L–R* circuit. What are its dimensions?

(Kanpur 2003, Rohilkhand 2004)

6. Discuss the charging and discharging of a capacitor through resistance. What is time constant?

(Gorakhpur 85, Rohilkhand 83)